Ji 2020 cutaneous SCC — analysis summary

RUN 3 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ=10 integrates 8 patients into one connected manifold; no per-sample islands.

Gate 2 — module geography → GO (branching)

Gast/pSom hub diverges into opposed Gast & pSom arms; Crest terminal on the pSom arm. A branching hub-and-spoke wishbone rooted at the Gast/pSom-high tip.

Gate 3 — hub cross-repression → GO (branching)

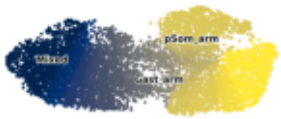
Mixed-hub Gast↔pSom mutual r = −0.47 (Puram −0.49); null 0.01±0.09, p<0.003.

STEP FIGURES

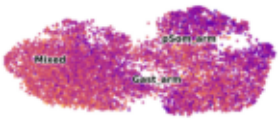
Manifold — samples



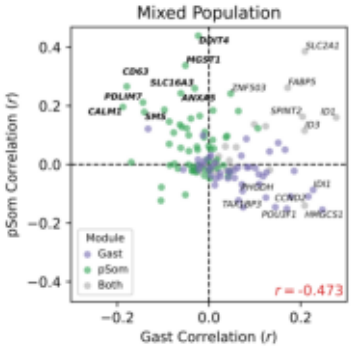
Trajectory / root



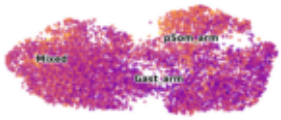
Gastrulation



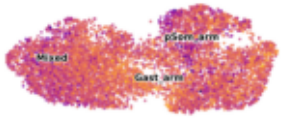
Gate 3: r = −0.47



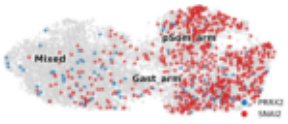
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous squamous cell carcinoma (Ji et al., 2020), 10x — 8/10 patients retained; 7,085 malignant developmental cells (10,531 pre-scope).

Verdict.
Embryonic branching pattern REPRODUCES — Gast/pSom hub, two cross-repressing arms, Crest terminal on the pSom arm (r=−0.47 vs Puram −0.49). Gate 1 GO / Gate 2 GO / Gate 3 GO. A Gast/pSom hub diverges into opposed Gast and pSom arms with the neural-crest terminus on the pSom arm; EMT-TFs PRRX2 (Gast arm) / SNAI2 (pSom arm terminus).

Parameter choices

Threshold	100 cells/sample (clean 6.70× count gap 40→268); keeps 8/10 patients (10,531 cells)	**Genes**	1,443 overdispersed (scFates); PCA 50 PC
Integration	Harmony θ=10 (10x precedent); branch survives strong integration	**Cycling scope**	3,446 cells in 5 high-cycling (>50%) Leiden lobes removed → 7,085-cell fork re-embedded (Filbin)
Embedding	kNN n_pcs=20 / n_neighbors=30; UMAP spread=1.3	**Trajectory**	scFates PPT tree Nodes=60 λ=3000 σ=0.5; 4 tips / 2 forks (hub-and-spoke Y)
Hub	fork-Leiden(res 0.4) max joint Gast+pSom = 'Mixed' (1,513 cells)	**Root**	Gast/pSom-high, Crest-low tip (root_score 0.99)
EMT-TFs	PRRX2 (Gast arm) / SNAI2 (pSom arm terminus)		